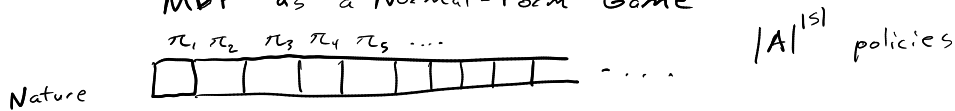


Learning in Games

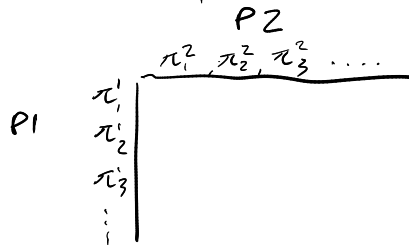
Markov Games, POSGs, Extensive Form

Dynamic Games

MDP as a Normal-Form Game



Dynamic Game in Normal Form



Classes of games where it's "easy" to find a NE

① Zero-sum

Optimization problem for finding NE

$$\begin{aligned} & \text{minimize}_{x, y, u, v} u - x^T A y + v - x^T B y \\ & \text{subject to} \begin{cases} u \geq x^T A y \\ v \geq x^T B y \end{cases} \quad \forall \text{ one-hot vectors } \hat{x} \begin{cases} \hat{x} \\ \hat{y} \end{cases} \left. \vphantom{\begin{matrix} u \geq x^T A y \\ v \geq x^T B y \end{matrix}} \right\} \text{Best response constraints} \\ & \sum_i x_i = 1 \\ & x_i \geq 0 \\ & \sum_i y_i = 1 \\ & y_i \geq 0 \end{aligned}$$

Zero-sum: $B = -A$

Solving a zero sum game is a LP

② Potential Games

$\exists \psi(a^1, a^2)$ defined so that if a player switches actions, their change in utility is equal to the change in ψ

↑ potential function

Examples:

Cooperative
Aligned

2, 2	1, 1
1, 1	3, 3

$A = B$

Prisoner's Dilemma

	coop	defect
coop	-4, -4	-10, -1
defect	-1, -10	-9, -9

$$\psi = \begin{bmatrix} 0 & 3 \\ 3 & 4 \end{bmatrix}$$

Iterative Best Response (IBR)
will find a NE in a potential game.

Any game can be decomposed into a potential game + zero-sum game

$$G = \underbrace{\bar{G}}_{\substack{\text{potential} \\ \text{easy}}} + \underbrace{G - \bar{G}}_{\substack{\text{zero-sum} \\ \text{easy}}}$$

↑
hard

↑
 $\frac{A+B}{2}$
easy

Why not just actor-critic?

- Policy Gradient may not converge (e.g. cycle)
- "Value function" depends on other player's policy, it is not easy to define in general-sum games

① Fictitious Play

Initialize $N^i(a) \leftarrow 0 \quad \forall i, a$

for $t \leftarrow 1$ to T

$\pi_t^i \leftarrow \text{normalize}(N^i)$ $\forall i \leftarrow$ distribution of best policies

$a^i \leftarrow \text{best_response}(\pi_t^{-i})$ $\vdots \leftarrow$ policy-gradient

$N^i(a^i) \leftarrow N^i(a^i) + 1$

return π_T

Does not always converge...

Often converges in practice!

Guaranteed convergence for 2-players:

- Constant sum
- Potential
- Nondegenerate $2 \times n$ (proven in 2005)
- Solvable by iterated elimination of strictly-dominated strategies

Example where it will not converge:

0,0	2,1	1,2
1,2	0,0	2,1
2,1	1,2	0,0

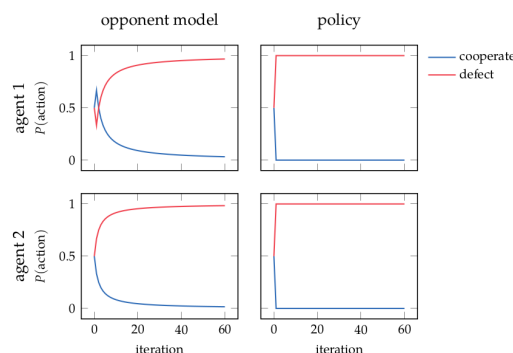


Figure 24.2. Two fictitious play agents learning and adapting to one another in a prisoner's dilemma game. The first row illustrates agent 1's learned model of 2 (left) and agent 1's policy (right) over iteration. The second row follows the same pattern, but for agent 2. To illustrate variation in learning behavior, the initial counts for each agent's model over the other agent's action were assigned to a random number between 1 and 10.

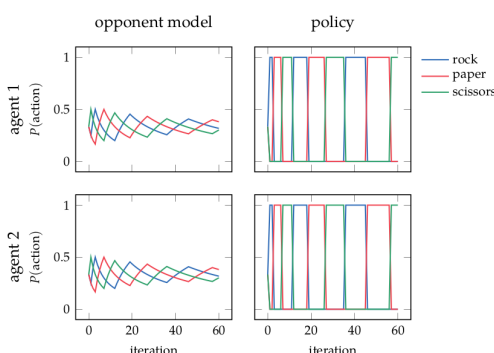


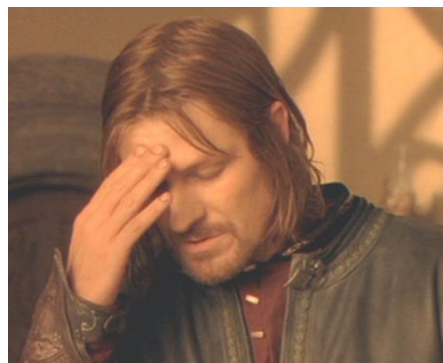
Figure 24.3. A visualization of two fictitious play agents learning and adapting to one another in a rock-paper-scissors game. The first row illustrates agent 1's learned model of 2 (left) and agent 1's policy (right) over time. The second row follows the same pattern, but for agent 2. To illustrate variation in learning behavior, the initial counts for each agent's action were assigned to a random number between 1 and 10. In this zero-sum game, fictitious play agents approach convergence to their stochastic policy Nash equilibrium.

② Regret Matching

Regret

How much better one could have done by taking another action.

$$R^i(a^i, \pi) = \underbrace{U^i(a^i, \pi^{-i})}_{\substack{\text{Not match} \\ H \quad T}} - \underbrace{U^i(\pi)}_{\text{Match}}$$



Match

H	1, -1	-1, 1
T	-1, 1	1, -1

$$\pi^1 = [0.5, 0.5] \quad \pi^2 = [0.5, 0.5] \quad \text{NE}$$

$$R^1(H, \pi) = (0.5 \cdot 1 + 0.5(-1)) - (0.25 \cdot 1 + 0.25(-1) + 0.25(1) + 0.25(-1)) = 0$$

$$R^1(T, \pi) = 0$$

$$R^2(H, \pi) = 0$$

$$R^2(T, \pi) = 0$$

Regret is 0 at NE!

$$\pi^1 = [1, 0]$$

$$\pi^2 = [1, 0]$$

$$R^1(H, \pi) = 1 - 1 = 0$$

$$R^1(T, \pi) = -1 - (-1) = 0$$

$$R^2(H, \pi) = 0$$

$$R^2(T, \pi) = (1) - (-1) = 2$$

Player 2 has a lot of regret.
They want to play T more

Regret Matching

a.k.a. Blackwell's Algo.

Initialize $\bar{\pi}_0 \leftarrow 0$ $\bar{R}_0^i(a) \leftarrow 0$ $\forall i, a$

for $t \in 1, T$

$$\pi_t^i \leftarrow \text{normalized}(\bar{R}_t^i) \quad \forall i$$

$$\bar{\pi}_t \leftarrow \bar{\pi}_{t-1} + \pi_t$$

$$R_t^i(a) \leftarrow \max(U^i(a, \pi_t^{-i}) - U^i(\pi_t), 0) \quad \forall i$$

$$\bar{R}_t^i \leftarrow \bar{R}_{t-1}^i + R_t^i \quad \forall i$$

return $\text{normalize}(\bar{\pi}_T)$

Informally:

Converges to a correlated equilibrium for any general-sum game.

(For a zero-sum game, a correlated equilibrium is a Nash Equilibrium)

"Counterfactual Regret" (CFR)

⇒ Regret matching at each information node in an extensive-form game